

Almodóvar Tufiño Tarek

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Summary

Undergraduate engineering student with a strong interest in software development, databases, and process automation.

Education

National Autonomous University of Mexico

B.S. in Computer Engineering

Expected Graduation: June 2027

Projects

Python AI Model

- Collaborated in a team to design and develop an AI chatbot model.
- Implemented search algorithms such as BFS (Breadth-First Search) and DFS (Depth-First Search) to optimize pathfinding in node-based maps.
- I improved the development of an BFS Algorithm that optimize the subways' travels by a 40% of the time.
- Integrated the Google Weather API to enable real-time weather responses from the agent.
- Utilized tree-based data structures to model the Mexico City Subway Network, improving route efficiency and navigation.

Autonomous Mini Sumo Robot (C++)

- Designed and programmed an autonomous robot using ESP32 tools, Arduino IDE and L298N motor driver.
 - Implemented obstacle detection using ultrasonic and infrared sensors.
 - Developed control logic for edge detection and opponent tracking.
 - Technologies: C/C++, Embedded Systems, Sensors.
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Experience

Currently working in the Instituto de Fisiología celular in UNAM in a Real-Time Neural Analysis Interface.

Technical Skills

Languages: Python, C/C++, Oracle SQL, MySQL, HTML, VHDL, CSS

Technologies Microsoft Office, Excel, PowerPoint, Word, Latex, AI Tools

Areas: Software Development, Algorithms, Data Analysis, Embedded Systems.

English: C1 (advanced)

Softskills

Teamwork, problem-solving, adaptability, willingness to learn.